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Interim Report on Investigations on Multi Disease Prediction in Dairy Cattle Using Machine Learning on French Dairy Cattle Dataset

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1 Background

2 Literature Review

One of the major achievements in the field of Machine Learning is the advancement in the area of disease predictions. These predictions are carried out with respect to the medical images, sensory data, or federated data. Among these medical predictions, the disease prediction of animals gained its attraction in the recent past. This is due to the fact that there are lots of regions like Austri, Switzerland, Ireland and so on where the majority of the investments lie in the area of agriculture and farming, where medical diagnosis of cattle is a serious problem. This is due to various factors such as : 1. availability of medical facilities, 2. Remote location of farms, 3. Awareness of farmers and so on. Therefore, it is inevitable for the agriculture industry to have a prediction system which can effectively classify and predict the cattle diseases. One of the successful algorithms developed in the recent past was deployed in the Russian Scientific Argo Engineering Centre, which detects the symptoms leading to many diseases such as nervous, infectious diseases and so on[1]. The studies reported by investigators faced many issues such as: 1. Shortage of data, 2. Biased data, generalizability of breeds and so on. Prio to the above study, Rober et al., 2020 developed a model for predicting the mastitis infection patterns in dairy cattle using SMOTE technique to develop synthetic data to adjust the imbalanced data [2]. Even though, the biasness in the minority cases were handled, the SMOTE technique did not improve the model much. Subsequently, Heidi et al., 2020 approached with an ensemble learning method to predict the dairy cattle survival and found out that regression ensemble performed better compared to others in the ensemble method[3]. This study was one of its kind were the regression ensemble predicted with a highest precision with AUC 0.561-0.731, followed by naives ensemble method. Following to this, adapting Apriori algorithm was yet another development brought out by Chandana et al., 2020 to help veterinarians diagnose infectious diseases I cattle from the electronic records[4]. It was interestingly found out that the algorithm performed better compared to SVM. However, as usual Imbalance, lower data quantity and generalizability is the drawback in the study. Apart from the above studies, a new algorithm was developed for the dataset involving not only cattle features but also environmental features [1]. In addition to the above, prediction of behavioural issues in Austrian Dairy Cows Under Subacute Ruminal Acidosis was attempted by Wagner et al., 2020[5]. In this study, characteristics of 28 cows were monitored for 9 weeks and found out that KNN gave the highest accuracy for detection. However, the lesser data content was the draw back. Following to this, with the same dataset, Wagner et al., investigated the reproductive diagnosis of cattle with respect to the daily activity of cows using BOSS and FBAT technique. [6]. Boss resulted in high speed , and FBAT gave high accuracy. However, generalizability and false alarm presence were the disadvantages of the study. Furthermore, in order to avoid the limitations in data quantity and quality, a foundation dataset was developed by Lardy et al., 2022 comprising of more than 300 cattle. This foundation dataset contains activity patterns and health patterns was cleaned, curated and ensured that it is viable for machine learning studies[7]. In this case, the absence of features such as the environmental features and 17.3 percent missing data were the drawbacks. Finally, the investigation by Lardy et al., 2023 utilized these 5 datasets together to classify diseases such as lameness, mastitis, acidosis, accident, oestrus, calving and so on[8]. The classification task aimed to predict the event rather

than disease and found out that reproductive event have the highest prediction accuracy accounting to 81In addition to the above disease prediction practices concerning the cattle, the ways in which the multiple predictions were carried out needs an extensive review to understand the potential possibilities. Various investigations reported the models on multiple predictions via. simultaneous multi output predictions and independent multi output prediction. The former focusses on predicting multiple outcomes together and the latter on predicting the multiple outcomes independently. The multilabel chromic disease prediction is one such study reported by Zhang et al., 2019 for the prediction of chromic diseases [9]. Groupnet, Googlenet, lSTM, SVM and DT were involved in the comparative analysis for the prediction of 110300 records with 62 features. Even though the Groupnet out performed other models, the associated computational cost and impbalance of the data adds to the drawback. Joint classification and regression using DM2L is the yet another style adopted by Liu et al., 2019 fo prediciting Alzheimers disease, where [10]. The approach analyzed MRI images and demographic information of patients including age, gender and education to predict. Even though the model is efficient, he computational cost is high and the important features such as genetical features which could improve the accuracy is absent. Later in 2021, another multilabel classification for medical purpose was attempted by Liu et al., 2021, for predicting protein subcellular localization[11]. The significant difference of this modelling from the other investigations is the incorporation of different feature extraction and dimensionality reduction techniques which makes this an advanced model. Apart from the disease prediction studies, the multilabel prediction is employed in NLP for classifying emotions from texts [12]. Multilabel KNN is the method used here which further divides the operations using sample and label based MLKNN for grammar feature in the adjacent sentences and the label correlation respectively. The main drawback with the model is the influence of label correlation on the computational cost and, lower performance associated with the small texts with single labels. Subsequently, hybrid multilabel model was introduced by Priyadarshini et al., 2023, the main aim of which was to compensate for the imbalance contributed by the minority data in datasets[13]. Synthetic data incorporation by adaptive synthetic data and ensemble learning. Despite the fact that the imbalance associated with the data is compensated, the scalability and generalizability associated with the synthetic data and computational cost adds to the cons of this technique. Apart from the above studies, the multi classification study by Gopisetti et al., 2023 predicted the presence of various diseases with respect to the patient data using various algorithms. The investigation achieved approximately 99 and 98 percent for chromic kidney and diabetes diseases respectively. The former was achieved with the Random Forest, while the later was achieved with the help of decision tree. Even though the model has got very good accuracy with potential deployment with streamlit, the unclear information on data imbalance, normalization, feature engineering and generalizability are unavailable.

Furthermore, the multi outputand class classification using gausian process was adopted by Chunchao et al., 2023 using a novel approach with linear model coregionalization and gumbel noise[15]. This advanced method classified each outputs as class and incorporates sub-sampling to reduce computational costs. Unlike other models, high scalability is the major highlight of this process. Whereas the feature loss during image down scaling and non viability in multi label classification are the cons of this work. This short coming was further investigated by Rahman et al., 2024 in his investigatins on breast cancer prediction using machine learning with the help of a multi output classifier [16].

3 Literature Review

One of the major achievements in the field of Machine Learning is the advancement in the area of disease predictions. These predictions are carried out with respect to medical images, sensory data, or federated data. Among these medical predictions, the disease prediction of animals has gained attention in the recent past. This is due to the fact that there are many regions like Austria, Switzerland, Ireland, and so on, where the majority of investments lie in agriculture and farming, and where medical diagnosis of cattle is a serious problem. This is due to various factors such as (1) availability of medical facilities, (2) remote location of farms, (3) awareness of farmers, and so on. Therefore, it is inevitable for the agriculture industry to have a prediction system that can effectively classify and predict cattle diseases.

One of the successful algorithms developed in the recent past was deployed in the Russian Scientific Agro Engineering Centre, which detects symptoms leading to many diseases such as nervous, infectious diseases, and so on [1]. The studies reported by investigators faced many issues such as: (1) shortage of data, (2) biased data, (3) generalizability of breeds, and so on. Prior to the above study, Rober et al., 2020 developed a model for predicting mastitis infection patterns in dairy cattle using the SMOTE (Synthetic Minoprity OverSampling Technique) technique to develop synthetic data to adjust for imbalanced data [2]. Even though the bias in the minority cases was handled, the SMOTE technique did not significantly improve the model.

Subsequently, Heidi et al., 2020 approached the problem with an ensemble learning method to predict dairy cattle survival and found that regression ensemble performed better compared to others in the ensemble method [3]. This study was one of its kind where the regression ensemble predicted with the highest precision with AUC (Are Under Curve)ranging from 0.561 to 0.731, followed by the naive ensemble method.

Following this, adapting the Apriori algorithm was another development introduced by Chandana et al., 2020 to help veterinarians diagnose infectious diseases in cattle from electronic records [4]. It was found that the algorithm performed better compared to SVM. However, as usual, imbalance, lower data quantity, and generalizability remained as drawbacks.

Apart from the above studies, a new algorithm was developed for a dataset involving not only cattle features but also environmental features [1]. In addition, prediction of behavioural issues in Austrian dairy cows under subacute ruminal acidosis was attempted by Wagner et al., 2020 [5]. In this study, characteristics of 28 cows were monitored for 9 weeks, and it was found that KNN gave the highest accuracy for detection. However, lesser data content was a drawback.

Following this, with the same dataset, Wagner et al. investigated the reproductive diagnosis of cattle with respect to the daily activity of cows using BOSS (Bag of Symbolic Fourier Approximation Symbols) and FBAT (Fourier Bases Approximation with Thresholding)techniques [6]. BOSS resulted in high speed, and FBAT gave high accuracy. However, generalizability and false alarm presence were the disadvantages of the study.

Furthermore, to avoid limitations in data quantity and quality, a foundation dataset was developed by Lardy et al., 2022 comprising more than 300 cattle [7]. This foundation dataset contains activity patterns and health patterns and was cleaned, curated, and ensured to be viable for machine learning studies. In this case, the absence of features

such as environmental data and 17.3 percent missing data were the drawbacks.

Finally, the investigation by Lardy et al., 2023 utilized these 5 datasets together to classify diseases such as lameness, mastitis, acidosis, accident, oestrus, calving, and so on [8]. The classification task aimed to predict the event rather than the disease and found that reproductive events had the highest prediction accuracy accounting for 81% using a Random Forest classifier compared to other classifiers.

In addition to the above disease prediction practices concerning cattle, the methods through which multiple predictions were carried out require an extensive review to understand potential possibilities. Various investigations reported models on multiple predictions via simultaneous multi-output predictions and independent multi-output prediction. The former focuses on predicting multiple outcomes together, and the latter on predicting multiple outcomes independently.

Multilabel chronic disease prediction is one such study reported by Zhang et al., 2019 for the prediction of chronic diseases [9]. GroupNet, GoogLeNet, LSTM (Lomg Short Term Memory), SVM (Support Vector Machine), and Decision Trees were involved in the comparative analysis for the prediction of 110,300 records with 62 features. Even though GroupNet outperformed other models, the associated computational cost and imbalance of the data were drawbacks.

Joint classification and regression using DM2L (Deep Multi task Multi channel Learning) is another approach adopted by Liu et al., 2019 for predicting Alzheimer's disease [10]. The approach analyzed MRI (Magnetic Resonance Imaging) images and demographic information of patients, including age, gender, and education. Even though the model was efficient, the computational cost was high, and important features such as genetic information, which could improve the accuracy, were absent.

Later in 2021, another multilabel classification for medical purposes was attempted by Liu et al., 2021, for predicting protein subcellular localization [11]. The significant difference in this model compared to other investigations was the incorporation of different feature extraction and dimensionality reduction techniques, which made it more advanced.

Apart from disease prediction studies, multi label prediction is also employed in NLP (Natural Language Processing) for classifying emotions from texts [12]. Multi label KNN (K Nearest Neighbor) is the method used here, which further divides the operations using sample- and label-based MLKNN (Multi Label K Nearest Neighbor) for grammar features in adjacent sentences and the label correlation, respectively. The main drawback of the model is the influence of label correlation on computational cost and lower performance associated with small texts with single labels.

Subsequently, a hybrid multilabel model was introduced by Priyadarshini et al., 2023, with the main aim of compensating for the imbalance contributed by the minority data in datasets [13]. This involved synthetic data incorporation by adaptive synthetic data and ensemble learning. Despite the fact that data imbalance was compensated, the scalability and generalizability of the synthetic data and computational cost added to the cons of this technique.

Apart from the above studies, the multi-classification study by Gopisetti et al., 2023 predicted the presence of various diseases based on patient data using various algorithms [14]. The investigation achieved approximately 99% and 98% accuracy for chronic kidney and diabetes diseases respectively. The former was achieved with the Random Forest, while the latter was achieved using the Decision Tree. Even though the model had high accuracy and potential deployment with Streamlit, information on data imbalance,

normalization, feature engineering, and generalizability was lacking.

Furthermore, multi-output and class classification using Gaussian processes was adopted by Chunchao et al., 2023 using a novel approach with linear model coregionalization and Gumbel noise [15]. This advanced method classified each output as a class and incorporated sub-sampling to reduce computational costs. Unlike other models, high scalability is the major highlight of this process. However, feature loss during image downscaling and non-viability in multilabel classification are the cons of this work.

This shortcoming was further investigated by Rahman et al., 2024 in his investigations on breast cancer prediction using machine learning with the help of a multi-output classifier [?].

4 Exploration of data and Methods

5 Proposed Future Analysis

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